

EXPERIMENT 7

AIM:

To explore random sampling and understand the concept of sampling distribution using Python.

CODE:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Step 1: Generate a population
```

```
population_mean = 100
```

```
population_std = 15
```

```
population_size = 50000
```

```
population = np.random.normal(  
    population_mean,  
    population_std,  
    population_size  
)
```

```
# Step 2: Random sampling
```

```
sample_sizes = [25, 50, 100]
```

```
num_samples = 500
```

```
sample_means = {}
```

```
for size in sample_sizes:
    sample_means[size] = [ ]

    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
            replace=False
        )
        sample_means[size].append(np.mean(sample))
```

Step 3: Plot sampling distributions

```
plt.figure(figsize=(12, 8))
```

```
for i, size in enumerate(sample_sizes):
```

```
    plt.subplot(3, 1, i + 1)
```

```
    plt.hist(
        sample_means[size],
        bins=30,
        alpha=0.7,
        label=f'Sample Size {size}'
    )
```

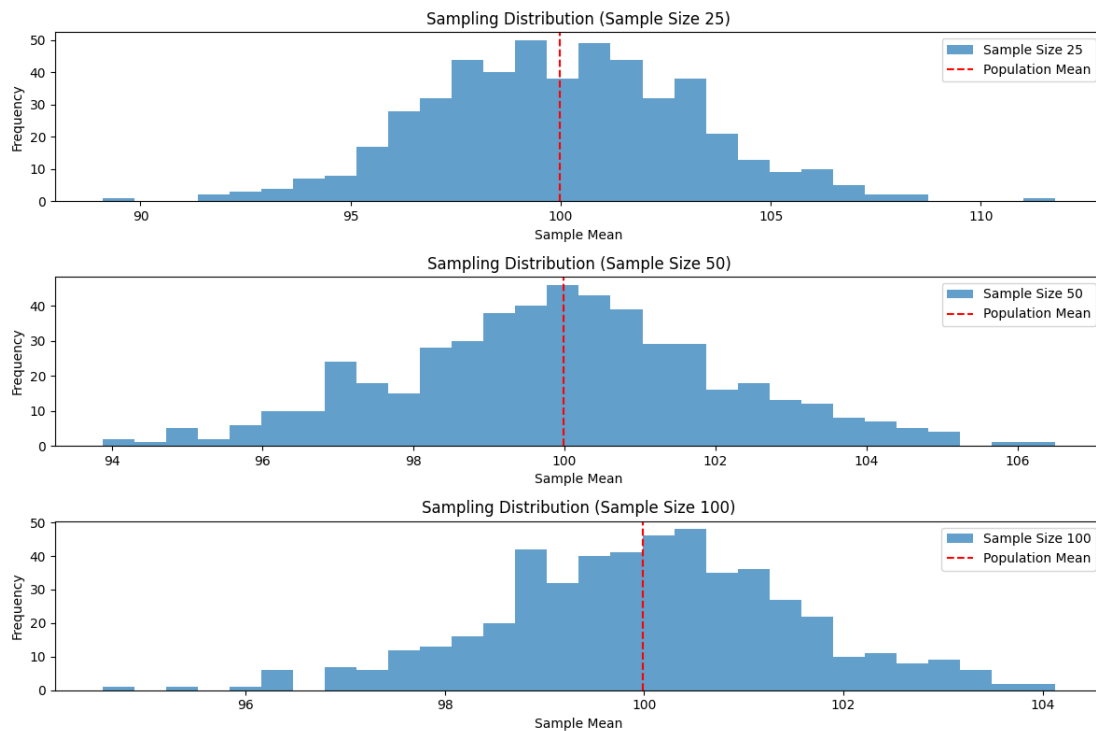
```
    plt.axvline(
```

```
np.mean(population),  
color='red',  
linestyle='--',  
linewidth=1.5,  
label='Population Mean'  
)
```

```
plt.title(f'Sampling Distribution (Sample Size {size})')  
plt.xlabel('Sample Mean')  
plt.ylabel('Frequency')  
plt.legend()
```

```
plt.tight_layout()  
plt.show()
```

OUTPUT:



RESULT:

Random samples and their sampling distributions were successfully generated. The results showed that larger sample sizes produce a narrower sampling distribution around the population mean.